

BBF202E Numerical Methods in Comp. Eng.

2025-2026 Spring - Term Project

Due: May 17, 2026

March 6, 2026

*“Question everybody, especially authority,
and choose your mutations carefully.”
Mark Mothersbaugh*

By turning in this assignment, I agree to the ITU honor code and declare that all of this is my own work.

Important Note

- Upload your solutions through Ninova. Homeworks sent via e-mail and late submissions will not be accepted.
- Please ensure that you include your full name and student identification number on every file you submit.
- Cheating is highly discouraged. It will be punished by a negative grade. Also, disciplinary actions will be taken. Please complete your homework on your own. Teamwork is not allowed. The pattern of your solutions must belong only to you.
- All code and reports will be subjected to plagiarism checks. Please do not copy any text or code from other sources.
- If you have any questions, please contact Res. Asst. Rumeysa Ashhan ERTÜRK (rumeysa.erturk@itu.edu.tr).

PCA-Based Ancestry Analysis Using SNPs

1 Project Overview

This project is based on the methodological framework introduced by Paschou et al. [1], which presents an unsupervised algorithm using Principal Components Analysis (PCA) and Singular Value Decomposition (SVD) to identify informative subsets of genetic markers. By selecting PCA-correlated SNPs that align with the top principal components, the method enables the reconstruction of genetic structure with up to 99% reduction in genotyping requirements. Notably, this approach detects population substructure without prior ancestry information and is effective for both broad and complex population analyses.

In this project, you will apply PCA and K-Nearest Neighbors (KNN) to analyze human genetic ancestry. Using SVD on high-dimensional SNP data, you will visualize population structure and identify key genetic markers by calculating SNP loading scores. You will then develop a classification model to predict the ancestral origins of test individuals based on their genetic proximity in PCA space, demonstrating the practical application of numerical methods in bioinformatics.

2 Background

A Single Nucleotide Polymorphism (SNP) is the most prevalent form of genetic variation in humans. It refers to a difference in a single nucleotide, which is one of the basic building blocks of DNA. For instance, a SNP may involve the substitution of cytosine (C) with thymine (T) at a specific position in the genome.

Approximately 99.9% of human DNA is identical among individuals. The remaining 0.1% includes SNPs, which are distributed throughout the genome. These variations are inherited from parents and transmitted across generations.

Although most SNPs are present in all human populations, the frequencies of particular SNPs differ substantially according to geographic and ancestral background. Such SNPs are known as Ancestry Informative Markers. Genetic drift and geographic isolation over thousands of years have led to certain genetic mutations becoming prevalent in specific populations, while remaining rare in others. For example, distinct SNP frequencies can be observed between populations in East Asia and Northern Europe.

In bioinformatics, SNP variations are represented numerically according to the number of alternate alleles present at a given locus.

- 0: Homozygous for the reference allele (e.g., CC)
- 1: Heterozygous (e.g., CT)
- 2: Homozygous for the alternate allele (e.g., TT)

By analyzing thousands of SNPs simultaneously, numerical methods such as PCA can be applied to identify patterns in genetic variation. Individuals with similar ancestral backgrounds exhibit comparable SNP profiles, resulting in clustering when visualized in two-dimensional space.

3 Project Description

3.1 Library Usage

The use of high-level library functions for core numerical methods—such as `np.linalg.eig`, `np.linalg.svd`, or scikit-learn's PCA—is strictly prohibited. While you are permitted to use basic NumPy arithmetic and matrix multiplication (`@` or `np.dot`), the implementation of the underlying numerical algorithms must be your own original work. Utilizing built-in solvers for these tasks will result in significant grade penalties.

3.2 Provided Data

The data for this project is structured into training and test sets, each consisting of genotype and label files. These files are provided in CSV format and serve as the foundation for the numerical analysis and classification tasks.

3.2.1 Genotype Data (`genotypes.csv` and `genotypes_test.csv`)

The genotype files represent the high-dimensional genetic profile of each individual.

- **Structure:** The data is organized as a matrix where each row represents an individual and each column represents a specific SNP.
- **Encoding:** Each cell contains a numeric value from the set $\{0, 1, 2\}$, representing the count of the alternative allele for that biallelic SNP (e.g., 0 for homozygous reference, 1 for heterozygous, and 2 for homozygous alternative).
- **Dimensionality:** These files contain thousands of columns (SNPs), creating a high-dimensional space that necessitates the use of PCA for dimensionality reduction.
- **Preprocessing:** Raw data may contain missing values (NaNs), which must be addressed through filtering or mean imputation before performing SVD.

3.2.2 Label Data (`labels.csv` and `labels_test.csv`)

The label files provide the “ground truth” for the individuals in the genotype files.

- **Columns:** Each file contains two primary columns: “IID” (Individual ID) and “POP” (Population).
- **Populations:** The individuals are assigned to one of four distinct ancestral populations, which serve as the target classes for the KNN classification model. In this project, we are using real population and SNP data collected from the HapMap3 project [2]. The examined populations in this project are:

- CHB: Han Chinese in Beijing, China,
- JPT: Japanese in Tokyo, Japan,
- TSI: Tuscans in Italy,
- YRI: Yoruba in Ibadan, Nigeria.

A fun (but unrelated) side note: For a long time, TSI was the standard genomic proxy for the Turkish population in global studies due to shared Mediterranean ancestry. However, recent large-scale sequencing has revealed that Turkish genomes are actually a unique “genetic bridge” with distinct West Asian and Central Asian layers.

3.2.3 Data Integration

To ensure a valid analysis, the genotype and label files must be correctly aligned.

- **Merging:** The scripts utilize the “IID” column to perform an inner merge, ensuring that only individuals with both genetic data and ancestry labels are included in the analysis.
- **Training vs. Test Split:** The training data is used to define the PCA space and SNP loading scores, while the test subjects are treated as “unknowns” to be projected into that established space for classification.

4 Project Tasks

TASK 1: Data Integration and Quality Control

- **Data Merging and Cleaning:** Load the genotype and label CSV files, merging them using the individual IDs (IID) to ensure each sample is correctly associated with its population label.
- **Missingness Filtering:** Implement a filter to remove SNPs and individuals with missing data (NaNs) exceeding specific thresholds (e.g., 5% for SNPs and 10% for individuals) to maintain data integrity.
- **Mean Imputation:** For any remaining missing genotypes, calculate the mean value per SNP from the available data and use it to fill the gaps, ensuring the final matrix contains only numerical values.

TASK 2: Implementing PCA

- **Matrix Centering:** Perform column-wise centering of the imputed genotype matrix by subtracting the mean of each SNP, which is a prerequisite for standard PCA.
- **SVD Calculation:** Use Singular Value Decomposition (SVD) to decompose the centered matrix into its singular values and eigenvectors (V matrix), which represent the principal directions of genetic variation.
- **Variance Analysis:** Compute the eigenvalues from the singular values to determine the variance explained by each Principal Component (PC).

TASK 3: Visualizing Genetic Structure

- **Scree and Cumulative Variance Plots:** Create plots to visualize the magnitude of eigenvalues and the cumulative variance ratio to determine how many PCs are necessary to represent the dataset structure.
- **2D Projection (PC Scatter Plots):** Project the individuals onto the first two PCs and color-code them by their population labels to observe how different ancestries cluster in the reduced space.

TASK 4: Feature Selection (SNP Scoring)

- **Calculating SNP Loading Scores:** Use the eigenvectors (V matrix) to calculate a score for each SNP based on its correlation with the top principal components.
- **Top SNP Identification:** Rank the SNPs by their scores and identify the top-K [10, 20, 50, 100, 200] SNPs that are most informative for defining the observed population structure. Create the PCA plots, show and save them on the notebook.

TASK 5: Ancestry Prediction using KNN

- **Test Data Projection:** Align the test genotype data with the training SNPs and project the test subjects into the PCA space created *with only 20 most descriptive SNPs* defined by the training set.
- **Implementing KNN Logic:** Apply a K-Nearest Neighbors (KNN) algorithm in the PCA space to classify the ancestry of test individuals based on their proximity to known training samples.
- **Performance Evaluation:** Compare the predicted labels against the true test labels using metrics such as overall accuracy and a confusion matrix to assess the model's reliability.

4.1 Provided Code

The project directory includes an IPython Notebook (.ipynb) that functions as the primary driver of the pipeline. This notebook contains a fully configured execution flow, with all required function calls arranged to support the analysis process.

The codebase is organized into multiple Python files, each acting as a library dedicated to a specific task, such as file input/output, data preprocessing, or PCA computations. Each script includes detailed documentation of the functions, specifying input and output arguments as well as their intended purposes.

Make sure that you complete the code for all functions marked with the comment `### STUDENT IMPLEMENTATION HERE ###`. The existing cells in the IPython Notebook should not be modified. Only execute the cells to monitor your progress. After the function bodies are implemented in the Python libraries, the notebook will automatically produce the necessary visualizations and generate log files for evaluation.

5 Submission

- Submit a zip file containing:
 - Python files under `src` folder
 - `main.ipynb` file under the main directory, the cells are executed, and cell outputs are shown on the notebook
 - Output log file and the figures under `out_dir` folder
- Submit a report summarizing your findings and experiments. Formatting Tips:
 - Use section headings (e.g., Introduction, Methods, Results, Discussion).
 - Label plots and tables clearly; caption them to highlight key findings.
 - Your report must adhere to the IEEE conference paper template in LaTeX, available at [this link](#).
 - Ensure proper citation formatting using IEEE citation style.
 - Maintain consistency in notation and terminology throughout your document.
 - Make sure that the report has at most 5 pages.

6 Assessment Criteria

The term project will be graded as follows:

- **80%: Implementation quality**
 - Implementation Correctness, Comparative Analysis, Visualization Clarity
- **20%: Report**
 - Organization, clarity, and adherence to guidelines.
 - Conciseness and completeness of the final write-up.

References

- [1] Peristera Paschou, Elad Ziv, Esteban G Burchard, Shweta Choudhry, William Rodriguez-Cintron, Michael W Mahoney, and Petros Drineas. Pca-correlated snps for structure identification in world-wide human populations. *PLoS genetics*, 3(9):e160, 2007.
- [2] The International HapMap 3 Consortium. Integrating common and rare genetic variation in diverse human populations. *Nature*, 467(7311):52–58, 2010.